Fiscal Year:	FY 2014	**Task Last Updated:**	FY 11/07/2014
PI Name:	Wyrobek, Andrew Ph.D.		
Project Title:	Molecular Characterization of Long-Term Degenerative CNS Risks from Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ajwyrobek@lbl.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	925-989-4466
Organization Name:	Lawrence Berkeley National Laboratory		
PI Address 1:	1 Cyclotron Rd MS: 74R0157		
PI Address 2:			
PI Web Page:			
City:	Berkeley	**State:**	CA
Zip Code:	94720-8099	**Congressional District:**	13
Comments:	For immediate assistance please contact my administrator Caron LaMarsh clamarsh@lbl.gov, 510.486.5317		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	01/01/2011	**End Date:**	06/30/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Loftus, David	**Contact Phone:**	650-604-1011
Contact Email:	david.j.loftus@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 6/30/2014 per A. Chu/NASA ARC (Ed., 4/10/14) NOTE: Extended to 3/31/2014 per A. Chu/NASA ARC (Ed., 12/16/13)		
Key Personnel Changes/Previous PI:	The key personnel on the project are Andrew J. Wyrobek (PI), Trent Northen (metabolomics), Ewa Witkowska (proteomics), Antoine Snijders (bioinformatics), Sandhya Bhatnagar (tissue biology).		
COI Name (Institution):	Snijders, Antoine (Lawrence Berkeley National Laboratory) Northen, Trent (Lawrence Berkeley National Laboratory) Witkowska, Ewa (University of California, San Francisco) Bhatnagar, Sandhya (Lawrence Berkeley National Laboratory) Straume, Tore (NASA, Ames) Rabin, Bernard (UMBC) Britten, Richard (UVMS)		
Grant/Contract No.:	NNA11AA58I		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objectives of this research are to characterize the persistent defects in neuro-signaling and cellular damage to central nervous system (CNS) tissue after cranial exposure to space radiation and to identify the critical molecular changes associated with individual variation in persistent neurocognitive deficits, using the rat model. We reported global changes in transcript expression profiles in the hippocampus and choroid plexus (CP) tissues of old animals that were exposed to 56Fe or 12C beams in their youth. We identified changes in the predicted protein expression in the CP that correlated with the magnitude of persistent neurocognitive deficit (i.e., to ~22 months of age). We are now conducting comparative analyses of radiation-induced changes in CP signaling and cellular architecture that are associated with neurocognitive deficits, using CNS tissue collected from animals exposed at various doses of 56Fe or 12C after they were tested for neurocognitive deficits. We are also investigating the early onset of radiation-induced changes to hippocampal and CP signaling, CP architecture, hippocampal neurogenesis, and composition of the cerebrospinal fluid (CSF) at ~3 months after exposure to both high and low-fluence 56Fe, and examining the association between molecular defects and neurocognitive changes within each irradiated animal. Our hypothesis is that radiation-induced damage to the CP can persist for the life of the individual, producing abnormal CSF that lacks critical metabolites and neurogenic factors needed for a normal CNS microenvironment, neurogenesis and cognitive functions. The Specific Aims are: Aim 1: Characterize the dose dependence of persistent signaling and cellular defects in the CP and hippocampal zones of neurogenesis of 12C- and 56Fe-exposed rats that have persistent deficits in their cognitive function and motor behavior. Aim 2: Characterize the time course of cellular and signaling defects in the CP of rats irradiated with 56Fe beams and determine the temporal relationship between CP defects and abnormal hippocampal neurogenesis. Aim 3: Determine whether irradiated animals with defective CP also produce CSF deficient in critical metabolites or proteins that are critical for adult neurogenesis, using proteomic and metabolomic approaches.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our project will provide a deeper understanding of the molecular and tissue mechanisms associated with individual variation in risks for neurocognitive deficits after exposure to space radiation. Specifically, our project will provide new information on: (a) the radiation response of the CP, a 3-gram organelle in humans located in the CNS ventricles, which is the major producer of CSF; (b) the molecular mechanisms affected by changes in signaling and damage to the cellular architecture of the CP; and (c) the quantitative relationship between the type and magnitude of radiation-induced damage to the CP and persistent neurocognitive dysfunctions in individuals exposed to space radiation. Our research utilizes a systems biology approach to investigate molecular signaling changes and cellular damage in CNS tissues in animals that show persistent neurocognitive declines after low-fluence HZE exposures. Understanding the genetic and physiological factors that determine the molecular signaling responses of the rodent brain after exposure to simulated space radiation is an important step towards reducing uncertainty in predicting risks for in-mission performance deficits and log-term risks of neurological diseases in astronauts on extended space missions.
	2014 FINAL REPORT PROGRESS (submitted November 2014) Our project provided NASA with an improved mechanistic understanding of the risks for impaired neurocognitive performance and individual susceptibility for neurological diseases after exposure to HZE radiation. Our approach used a rat model and: (a) employed state-of-the-art ‘omics and tissue technologies to investigate CNS damage response functions in animals exposed to low-fluence HZE particles (i.e., <1 hit per cell), (b) linked the type and magnitude of molecular defects with neurocognitive performance, and (c) validated the predicted damage response functions in brain tissue. Our project focused on CNS tissue damage to the hippocampus (site of spatial and performance memories) and the choroid plexus (CP, small cauliflower-like tissue in the ventricles adjacent to the hippocampus that produces cerebral spinal fluid (CSF) and maintains blood-brain homeostasis). Our research distinguished between two broad categories of risk mechanisms: (1) mechanisms related to exposure dose, and (2) mechanisms of inter-individual variations in risks for neurocognitive performance, and neurological diseases (e.g., Alzheimer Disease (AD)). Our working model is that neurocognitive damage after radiation involves direct cell damage to specific CNS target cells as well as indirect damage mediated by the irradiated microenvironment, such as abnormal CSF produced by an abnormal choroid plexus. The hypothesis was that exposure to space-relevant doses of HZE radiation (less than one ion per cell) induces abnormal molecular signaling and cellular damage in CP and hippocampus that determine the risks and individual susceptibilities for neurocognitive deficits and neurological diseases such as AD. We confirmed the radiation-sensitivity of the CP, measuring significant cellular and molecular damage to the CP after exposure to X-ray and 56Fe radiation. Our experiments include young and age-matched old reference groups to provide a means to distinguish between radiation effects and normal aging. We discovered long-lived molecular signaling changes in the CNS after exposure to space radiation using transcriptomic, proteomic and metabolomics analyses of several brain regions (hippocampus, CP, CSF, and cortex) collected at ~3 days to ~21 months after whole body HZE or X-ray exposures (10-200cGy), usually for rats also tested on one or more psychometric assay. Expression profiling and bioinformatics identified a significant Huntington’s disease subnetwork and significant associations with neuroinflammation, oxidative stress, and various neurological diseases and neurocognitive dysfunctions. We demonstrated that cranial exposure of young adult rats to heavy particle ionizing radiation (HZE, 56Fe, and 12C) causes neurocognitive deficits that persist to very old age, that the magnitude of the deficits are variable across animals and was associated with molecular and cellular damage to the CP and hippocampus. At ~2 years after exposures, there were histological changes of the CP including occluded endothelial vessels, increases in the frequency of amyloid deposits throughout the CNS, and decreased epithelial expression of proteins that normally prevent amyloid fibril formation in CNS. In support of observation, we found increased frequencies of amyloid plaques in exposed animals but noted that the increase in plaque frequency was delayed 4-9 months after exposure. Our findings suggest that space radiation may increase the risks for Alzheimer-related pathologies in the brain, long after the exposure. We also confirmed the hypothesis that radiation exposure damaged the ability of the CP to produce normal cerebral spinal fluid. Animals exposed to 10 and 100cGy 56Fe showed changes in many proteins associated with various neurological diseases, psychological disorders, neuronal cell death and survival. We also discovered molecular signaling changes that

Task Progress:	are associated with inter-individual variation in behavioral responses to HZE radiation, related to memory deficits and increased anxiety in rats. In summary, our research in a rat model identified persistent histological and molecular defects in the various CNS tissues (CP, CSF, and hippocampus) that are different from normal aging and are associated with long-term neurocognitive dysfunction related to memory deficits, increased anxiety, and increased plaque formation long after HZE exposures. 2013 TASK PROGRESS (Reported November 2013) This annual report covers the 3rd year of our 3-year project. We conducted seven major whole body radiation experiments in a rat model and collected CNS tissue for transcriptomic, proteomic, metabolomics, and for in situ analyses of CNS brain regions (hippocampus, choroid plexus, CSF, and cortex). Tissues were collected from 3 days to 21 months after whole body HZE or X-ray exposures (10-200cGy), usually for animals that were also tested on one or more psychometric assay. We employed advanced 'omic tools and in situ methods to (1) investigate molecular changes in brain tissue regions of rats exposed to simulated space radiation with emphasis on low dose exposures, and (2) examine the association between molecular changes, tissue damage, and psychometric outcomes. A major technical accomplishment was the development of a reliable method for collecting cerebral spinal fluid (CSF) from live animals for proteomic analyses, and the ability to isolate the fragile choroid plexus (CP) tissue for metabolomics and proteomic analyses. We have identified radiation-response signaling mechanisms in the choroid plexus (CP) and hippocampus as well as mechanisms associated with individual variations in the severity and persistence of CP tissue pathology and neurocognitive deficits after exposures to space radiation. Using transcript profiling in the rat model, we discovered persistent hippocampal and cortical signaling changes at ~22 months after exposure to 56Fe (100cGy) or 12C (10, 100, 150, 200cGy) in animals that also showed longterm neurocognitive deficits when compared with age-matched shams and 2-month-old animals. At the higher doses, the hippocampal and cortical regions showed significantly modulated genes associated with inflammatory response and several neurological diseases and disorders (Alzheimer, Parkinson, schizophrenia, etc.). 56Fe-exposed rats showed a down-regulated subnetwork of genes associated with Huntington disease. The lowest dose tested in this study (10cGy 12C) induced persistent behavioral deficits along with changes in immune and neuronal signaling in the hippocampus. In situ analyses of the second hemisphere from the same animals showed damage to the CP architecture and expression changes beyond what was observed with aging alone. Intraluminal endothelial occlusions were broadly evident within both the CP and hippocampus of irradiated animals and the occlusions stained positive for fibrinogen, keratin, and other proteins. Consistent with the transcript findings, transthyretin (ttr) protein expression was significantly reduced in the CP epithelium of both beam types. Using the inter-animal variation to advantage, we observed significant correlations between ttr transcript expression and neurocognitive test scores across animals. Ttr is known to prevent Ab amyloid fibril formation, a key pathological feature in Alzheimer's disease. Thus, these findings point to the CP organelle as a major target organelle for persistent radiation defects associated with neurocognitive function. Then, we hypothesized that a radiation-damaged CP might produce cerebral spinal fluid (CSF) that is less capable of supporting normal adult neurogenesis and neurocognitive function. We collected CSF from middle-aged rats at 2-3 months after irradiation for analyses of CSF proteomic along with CP tissue for IHC. CSF proteomics identified ~400 distinct rat proteins, of which ~45 showed changes with aging (2 months vs 9 months). The animals exposed to 10 and 100cGy animals showed changes in 24 and 26 proteins that were significantly associated with neurological disease, psychological disorders, cell death and survival. These findings support the hypothesis that the CP of irradiated animals produced CSF containing changes in proteins associated with neurocognitive diseases and disorders. Studies are ongoing to characterize the cellular in situ expression of radiation-sensitive proteins in the CP and hippocampus of the same irradiated animals that provided the CSF for proteomics to identify and characterize candidate mechanism underlying radiation-induced behavioral dysfunctions. These studies utilize larger groups of out bred animals that were categorized according to their behavioral responses (memory and anxiety) after exposure to the same low doses of ionizing radiation, to identify the molecular mechanisms for differences in individual susceptibility to behavioral deficits after common exposures to space radiation. Understanding the genetic and physiological factors that determine the molecular signaling responses of the rodent brain after exposure to simulated space radiation is an important step towards reducing uncertainty in predicting risks for in-mission performance deficits and log-term risks of neurological diseases in astronauts on extended space missions.
Bibliography Type:	Description: (Last Updated: 06/19/2025)
Abstracts for Journals and Proceedings	Wyrobek AJ, Lowe XR, Rabin BM, Bhatnager S, Rademaker P, Northen T, Witkowska E, Straume T, Snijders A. "Molecular defects in the choroid plexus and hippocampus of animals with neurocognitive deficits after exposure to space radiation." 24th Annual NASA Space Radiation Investigators' Workshop at HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. Presentation. 24th Annual NASA Space Radiation Investigators' Workshop at HITSRS2013--Heavy Ion in Therapy and Space Radiation Symposium 2013, Chiba, Japan, May 15-18, 2013. , May-2013
Articles in Peer-reviewed Journals	Lowe XL, Wyrobek AJ. "Characterization of the early CNS stress biomarkers and profiles associated with neuropsychiatric diseases." Current Genomics. 2012 Sep;13(6):489-97. http://dx.doi.org/10.2174/138920212802510448 ; PubMed PMID: 23449133 , Sep-2012